

VLSI Devices

Lecture 24

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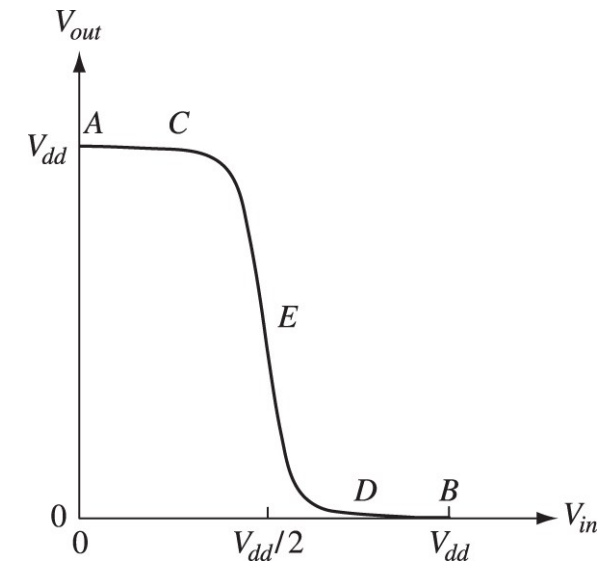
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3D structure of a FinFET inverter

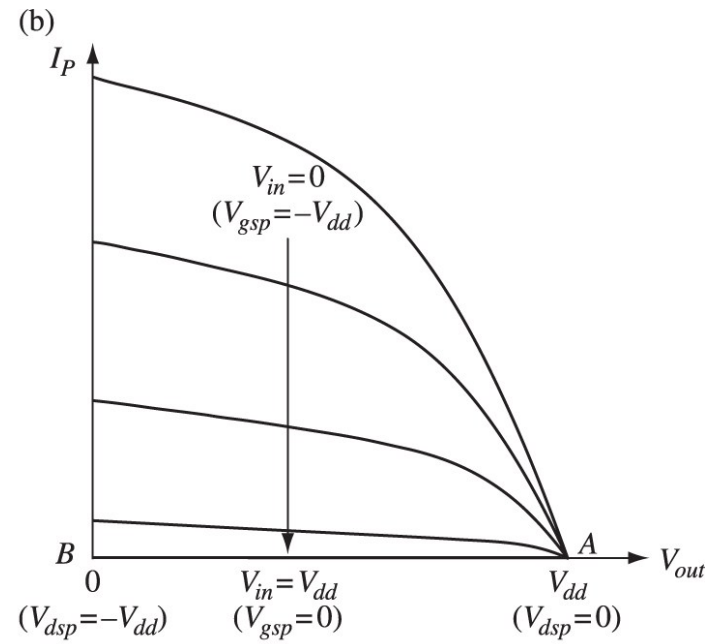
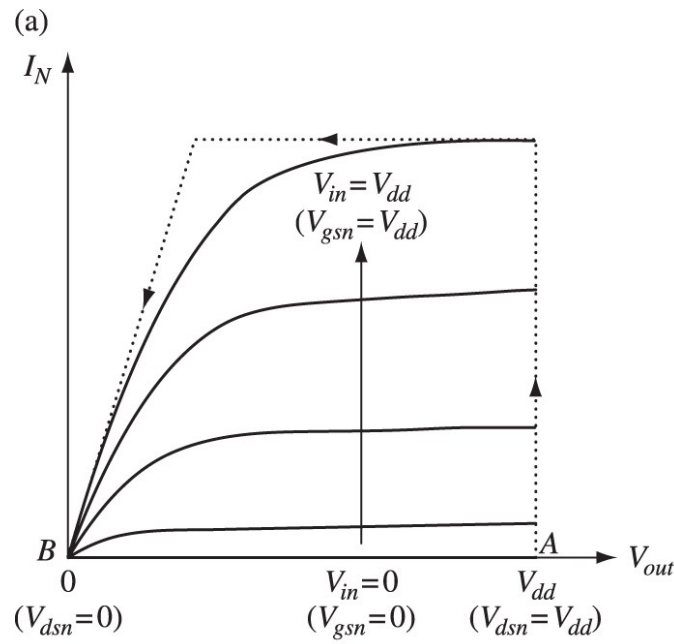
- A generic technology (CPP = 54 nm)
 - FEOL
 - MOL
 - BEOL

CMOS inverter transfer curve

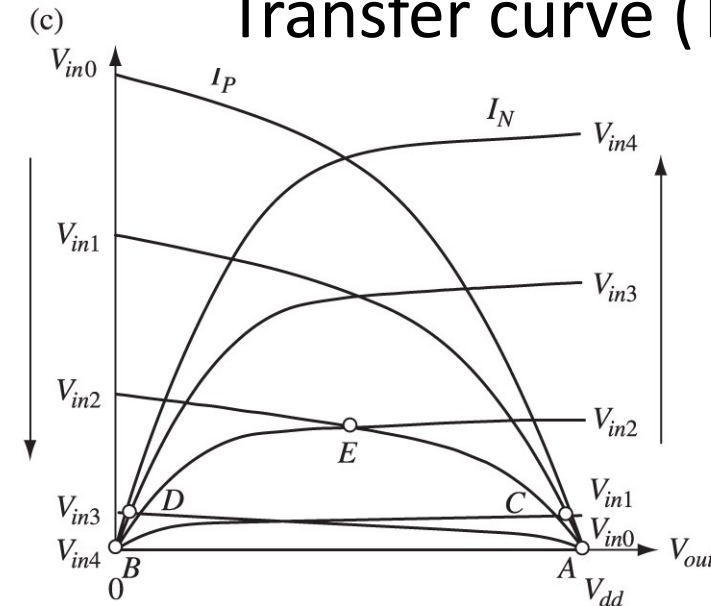
- High-to-low transition near the midpoint
 - It is desirable for I_P and I_N to be nearly symmetrical.



Transfer curve (Taur, Fig. 8.5)

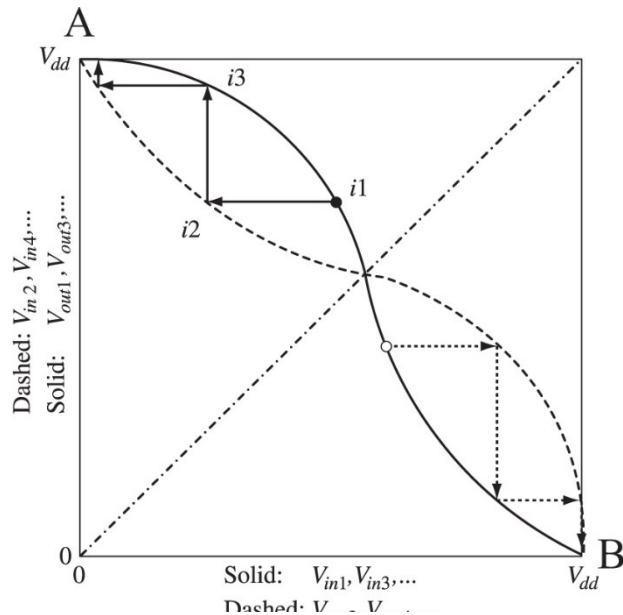


IV curves of transistors (Taur, Fig. 8.4)

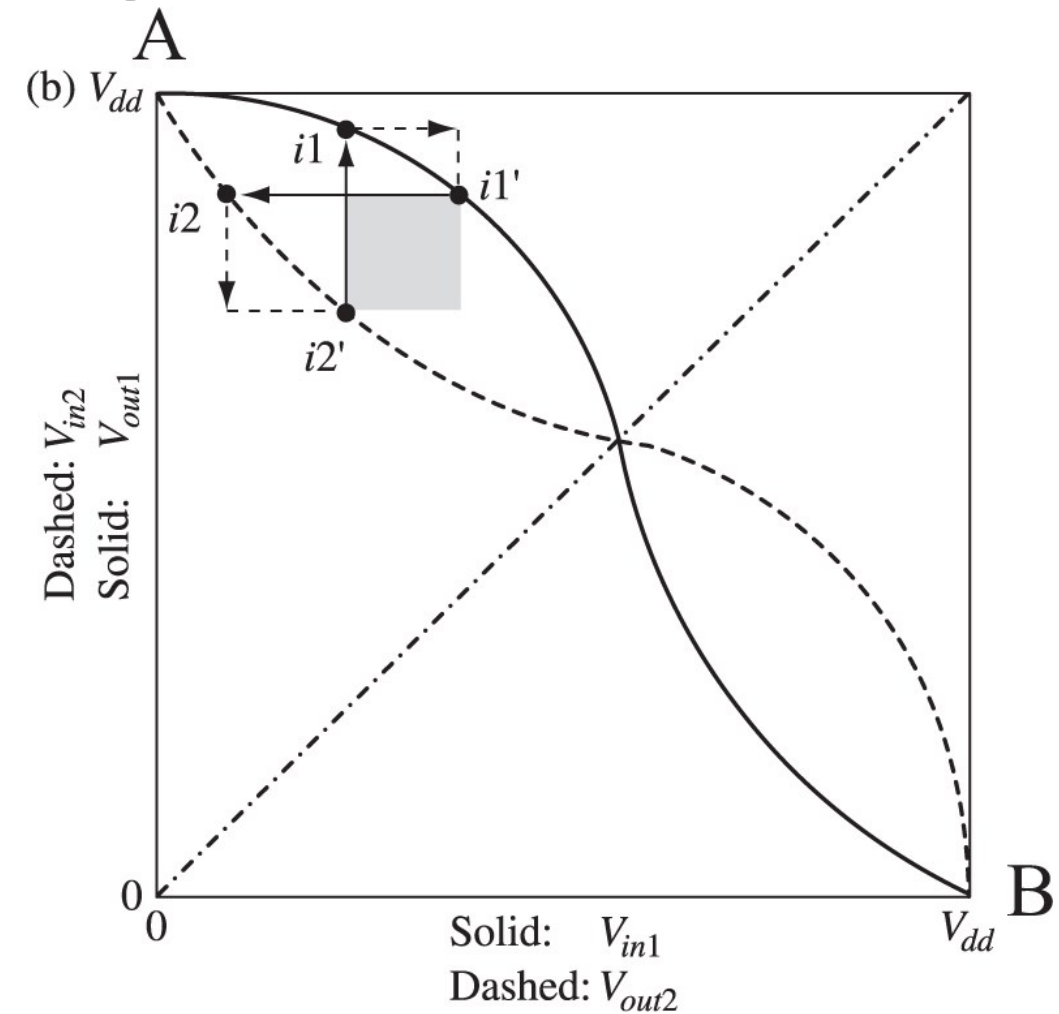


CMOS inverter noise margin

- Worst-case scenario
 - At each stage, we add a constant noise voltage.



Signal recovery by an inverter chain
(Taur, Fig. 8.7)



Noise margin (Taur, Fig. 8.8(b))

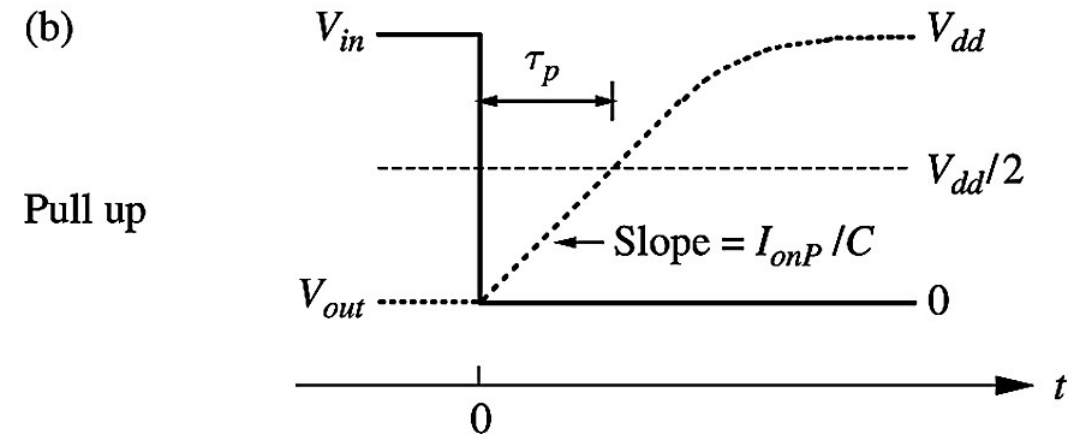
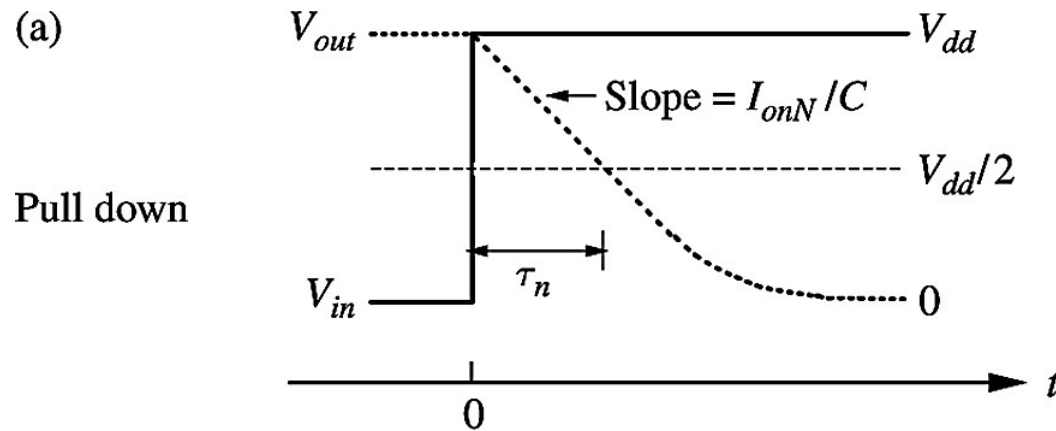
CMOS inverter delay

- Pull-down delay (τ_n) and pull-up delay (τ_p)

– Time delay to reach $\frac{V_{dd}}{2}$

$$\tau_n = \frac{CV_{dd}}{2I_{onN}}$$

Taur, Eq. (8.8)



Waveform of the output node voltage (Taur, Fig. 8.9)

Switching power dissipation

- An energy of CV_{dd}^2 is dissipated per cycle.
 - When the period is T ,

$$P = \frac{CV_{dd}^2}{T} = CV_{dd}^2 f$$

Taur, Eq. (8.10)

Thank you!